

Montserrat Nafria

其他

基本信息

姓名: Montserrat Nafria

所属院校: 巴塞罗那自治大学

地区: Spain

企业经历: 有

学科领域

微电子科学与工程

材料物理

集成电路设计与集成系统

研究领域

微电子器件与制造

薄膜材料与表面物理

纳米电子与先进工艺

微电子封装与可靠性

材料微结构与表征

EDA工具与电路验证

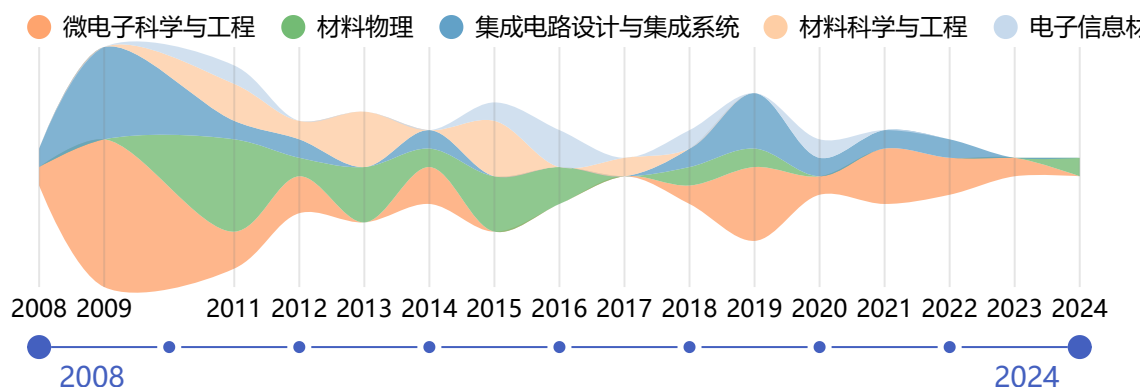
纳米物理与纳米器件

材料缺陷与微结构分析

个人简介

Montserrat Nafria is an academic researcher from the Autonomous University of Barcelona. She has contributed to research on topics such as Conductive atomic force microscopy and Gate oxide. With an h - index of 32, she has co - authored 276 publications. Her previous affiliations include SEMATECH and Katholieke Universiteit Leuven. Her research work involves areas like resistive memory switching mechanisms and the role of grain boundaries in resistive switching.

近年发表的论文



On the Aging of OTFTs and Its Impact on PUFs Reliability

MICROMACHINES

Porti, Marc; Palau, Gerard; Crespo-Yepes, Albert; Rus, August Arnal; Ogier, Simon; Ramon, Eloi; Nafria, Montserrat

被引用次数: 0

2024

Oxide Breakdown Spot Spatial Patterns as Fingerprints for Optical Physical Unclonable Functions

IEEE ELECTRON DEVICE LETTERS

Porti, Marc; Redon, Miquel; Munoz, Jordi; Nafria, Montserrat; Miranda, Enrique

被引用次数: 0

2023

Beneficial Role of Noise in Hf-based Memristors

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Rodriguez, Rosana; Martin-Martinez, Javier; Salvador, Emili; Crespo-Yepes, Albert; Miranda, Enrique; Nafria, Montserrat; Rubio, Antonio; Ntinis, Vasileios; Sirakoulis, Georgios Ch.

被引用次数: 4

2022

On the Impact of the Biasing History on the Characterization of Random Telegraph Noise

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Saraza-Canflanca, Pablo; Castro-Lopez, Rafael; Roca, Elisenda; Martin-Martinez, Javier; Rodriguez, Rosana; Nafria, Montserrat; Fernandez, Francisco, V

被引用次数: 0

2022

Aging in CMOS RF Linear Power Amplifiers: An Experimental Study

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Aragones, Xavier; Barajas, Enrique; Crespo-Yepes, Albert; Mateo, Diego; Rodriguez, Rosana; Martin-Martinez, Javier; Nafria, Montserrat

被引用次数: 13

2021

Power-Efficient Noise-Induced Reduction of ReRAM Cell's Temporal Variability Effects

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ntinis, Vasileios; Rubio, Antonio; Sirakoulis, Georgios Ch; Aguilera, Emili Salvador; Pedro, Marta; Crespo-Yepes, Albert; Martin-Martinez, Javier; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 16

2021

Statistical threshold voltage shifts caused by BTI and HCI at nominal and accelerated conditions

SOLID-STATE ELECTRONICS

Diaz-Fortuny, Javier; Saraza-Canflanca, Pablo; Rodriguez, Rosana; Martin-Martinez, Javier; Castro-Lopez, Rafael; Roca, Elisenda; V. Fernandez, Francisco; Nafria, Montserrat

被引用次数: 4

2021

Exploring the resistance change per energy unit as universal performance parameter for resistive switching devices

SOLID-STATE ELECTRONICS

Gomez, Jorge; Vourkas, Ioannis; Abusleme, Angel; Rodriguez, Rosana; Martin-Martinez, Javier; Nafria, Montserrat; Rubio, Antonio

被引用次数: 7

2020

Flexible Setup for the Measurement of CMOS Time-Dependent Variability With Array-Based Integrated Circuits

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Diaz-Fortuny, Javier; Saraza-Canflanca, Pablo; Castro-Lopez, Rafael; Roca, Elisenda; Martin-Martinez, Javier; Rodriguez, Rosana; Fernandez, Francisco V.; Nafria, Montserrat

被引用次数: 16

2020

DC Characterization and Fast Small-Signal Parameter Extraction of Organic Thin Film Transistors With Different Geometries

IEEE ELECTRON DEVICE LETTERS

Arnal, August; Crespo-Yepes, Albert; Ramon, Eloi; Teres, Lluís; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 5

2020

A smart noise- and RTN-removal method for parameter extraction of CMOS aging compact models

SOLID-STATE ELECTRONICS

Diaz-Fortuny, Javier; Martin-Martinez, Javier; Rodriguez, Rosana; Castro-Lopez, Rafael; Roca, Elisenda; Fernandez, Francisco, V.; Nafria, Montserrat

被引用次数: 4

2019

Low-Power, High-Performance, Non-volatile Inkjet-Printed HfO₂-Based Resistive Random Access Memory: From Device to Nanoscale Characterization

ACS APPLIED MATERIALS & INTERFACES

Vescio, Giovanni; Martin, Gemma; Crespo-Yepes, Albert; Claramunt, Sergi; Alonso, Daniel; Lopez-Vidrier, Julian; Estrade, Sonia; Porti, Marc; Rodriguez, Rosana; Peiro, Francesca; Cornet, Albert; Cirera, Albert; Nafria, Montserrat

被引用次数: 18

2019

Self-Organizing Neural Networks Based on OxRAM Devices under a Fully Unsupervised Training Scheme

MATERIALS

Pedro, Marta; Martin-Martinez, Javier; Maestro-Izquierdo, Marcos; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 10

2019

Experimental Verification of Memristor-Based Material Implication NAND Operation

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Maestro-Izquierdo, Marcos; Martin-Martinez, Javier; Crespo Yepes, Albert; Escudero, Manel; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Rubio, Antonio

被引用次数: 3

2019

A Versatile CMOS Transistor Array IC for the Statistical Characterization of Time-Zero Variability, RTN, BTI, and HCI

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Diaz-Fortuny, Javier; Martin-Martinez, Javier; Rodriguez, Rosana; Castro-Lopez, Rafael; Roca, Elisenda; Aragonés, Xavier; Barajas, Enrique; Mateo, Diego; Fernandez, Francisco V.; Nafria, Montserrat

被引用次数: 34

2019

Performance and Power Consumption Trade-Off in UTBB FDSOI Inverters Operated at NTV for IoT Applications

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Couso, Carlos; Martin-Martinez, Javier; Porti, Marc; Nafria, Montserrat

被引用次数: 5

2018

Editorial to J-EDS ESSDERC 2017 Special Issue

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Nafria, Montserrat; Collaert, Nadine

被引用次数: 0

2018

Experimental Time Evolution Study of the HfO₂-Based IMPLY Gate Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Maestro-Izquierdo, Marcos; Martin-Martinez, Javier; Yepes, Albert Crespo; Escudero, Manel; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Rubio, Antonio

被引用次数: 6

2018

Protective nanometer films for reliable Cu-Cu connections

MICROELECTRONICS RELIABILITY

Berthold, Tobias; Benstetter, Guenther; Frammelsberger, Werner; Bogner, Manuel; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 0

2017

Evaluation of ultra-thin structures composed of graphene and high-k dielectrics for resistive switching memory applications

INTERNATIONAL JOURNAL OF NANOTECHNOLOGY

Wu, Qian; Claramunt, Sergi; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 0

2016

Investigation of Filamentary Current Fluctuations Features in the High-Resistance State of Ni/HfO₂-Based RRAM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bargallo Gonzalez, Mireia; Martin-Martinez, Javier; Maestro, Marcos; Cruz Acero, Maria; Nafria, Montserrat; Campabadal, Francesca

被引用次数: 25

2016

Nanoscale characterization of copper oxide films by Kelvin Probe Force Microscopy

THIN SOLID FILMS

Berthold, Tobias; Benstetter, Guenther; Frammelsberger, Werner; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 11

2015

Nanoscale characterization of CH₃-terminated Self-Assembled Monolayer on copper by advanced scanning probe microscopy techniques

APPLIED SURFACE SCIENCE

Berthold, Tobias; Benstetter, Guenther; Frammelsberger, Werner; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 4

2015

In Situ Demonstration of the Link Between Mechanical Strength and Resistive Switching in Resistive Random-Access Memories

ADVANCED ELECTRONIC MATERIALS

Shi, Yuanyuan; Ji, Yanfeng; Hui, Fei; Nafria, Montserrat; Porti, Marc; Bersuker, Gennadi; Lanza, Mario

被引用次数: 12

2015

Conductive-AFM topography and current maps simulator for the study of polycrystalline high-k dielectrics

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Couso, Carlos; Porti, Marc; Martin-Martinez, Javier; Iglesias, Vanessa; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 2

2015

New Weighted Time Lag Method for the Analysis of Random Telegraph Signals

IEEE ELECTRON DEVICE LETTERS

Martin-Martinez, Javier; Diaz, Javier; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 52

2014

Analysis of Factors in the Nanoscale Physical and Electrical Characterization of High-K Materials by Conductive Atomic Force Microscope

INTEGRATED FERROELECTRICS

Zhang, Kai; Lanza, Mario; Shen, Ziyong; Fu, Qiang; Hou, Shimin; Porti, Marc; Nafria, Montserrat

被引用次数: 1

2014

A Conductive AFM Nanoscale Analysis of NBTI and Channel Hot-Carrier Degradation in MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wu, Qian; Bayerl, Albin; Porti, Marc; Martin-Martinez, Javier; Lanza, Mario; Rodriguez, Rosana; Velayudhan, Vikas; Nafria, Montserrat; Aymerich, Xavier; Bargallo Gonzalez, Mireia; Simoen, Eddy

被引用次数: 6

2014

Leakage current through the poly-crystalline HfO₂: Trap densities at grains and grain boundaries

JOURNAL OF APPLIED PHYSICS

Pirrotta, Onofrio; Larcher, Luca; Lanza, Mario; Padovani, Andrea; Porti, Marc; Nafria, Montserrat; Bersuker, Gennadi

被引用次数: 84

2013

Nanoscale and device level electrical behavior of annealed ALD Hf-based gate oxide stacks grown with different precursors

MICROELECTRONICS RELIABILITY

Bayerl, Albin; Lanza, Mario; Aguilera, Lidia; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; de Gendt, Stefan

被引用次数: 8

2013

Electrical and mechanical performance of graphene sheets exposed to oxidative environments

NANO RESEARCH

Lanza, Mario; Wang, Yan; Gao, Teng; Bayerl, Albin; Porti, Marc; Nafria, Montserrat; Zhou, Yangbo; Jing, Guangyin; Zhang, Yanfeng; Liu, Zhongfan; Yu, Dapeng; Duan, Huiling

被引用次数: 37

2013

Circuit Design-Oriented Stochastic Piecewise Modeling of the Postbreakdown Gate Current in MOSFETs: Application to Ring Oscillators

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Martin-Martinez, Javier; Kaczer, Ben; Degraeve, Robin; Roussel, Philippe J.; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Dierickx, B.; Groeseneken, Guido

被引用次数: 9

2012

Resistive switching in hafnium dioxide layers: Local phenomenon at grain boundaries

APPLIED PHYSICS LETTERS

Lanza, M.; Bersuker, G.; Porti, M.; Miranda, E.; Nafria, M.; Aymerich, X.

被引用次数: 158

2012

Grain boundaries as preferential sites for resistive switching in the HfO₂ resistive random access memory structures

APPLIED PHYSICS LETTERS

Lanza, M.; Zhang, K.; Porti, M.; Nafria, M.; Shen, Z. Y.; Liu, L. F.; Kang, J. F.; Gilmer, D.; Bersuker, G.

被引用次数: 184

2012

Injected Charge to Recovery as a Parameter to Characterize the Breakdown Reversibility of Ultrathin HfSiON Gate Dielectric

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Crespo-Yepes, Albert; Martin-Martinez, Javier; Rothschild, Aude; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 3

2011

Gate Voltage Influence on the Channel Hot-Carrier Degradation of High-k-Based Devices

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Amat, Esteve; Kauerauf, Thomas; Degraeve, Robin; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Groeseneken, Guido

被引用次数: 13

2011

Nanoscale and Device Level Gate Conduction Variability of High-k Dielectrics-Based Metal-Oxide-Semiconductor Structures

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Bayerl, Albin; Lanza, Mario; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; Campabadal, F.; Benstetter, Guenther

被引用次数: 11

2011

Metal oxide resistive memory switching mechanism based on conductive filament properties

JOURNAL OF APPLIED PHYSICS

Bersuker, G.; Gilmer, D. C.; Veksler, D.; Kirsch, P.; Vandelli, L.; Padovani, A.; Larcher, L.; McKenna, K.; Shluger, A.; Iglesias, V.; Porti, M.; Nafria, M.

被引用次数: 456

2011

Conductivity and Charge Trapping After Electrical Stress in Amorphous and Polycrystalline Al₂O₃-Based Devices Studied With AFM-Related Techniques

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Lanza, Mario; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; Benstetter, Guenther; Lodermeier, Edgar; Ranzinger, Heiko; Jaschke, Gert; Teichert, Steffen; Wilde, Lutz; Michalowski, Pawel Piotr

被引用次数: 28

2011

Grain boundary-driven leakage path formation in HfO₂ dielectrics

SOLID-STATE ELECTRONICS

Bersuker, G.; Yum, J.; Vandelli, L.; Padovani, A.; Larcher, L.; Iglesias, V.; Porti, M.; Nafria, M.; McKenna, K.; Shluger, A.; Kirsch, P.; Jammy, R.

被引用次数: 112

2011

Combined Nanoscale and Device-Level Degradation Analysis of SiO₂ Layers of MOS Nonvolatile Memory Devices

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Lanza, Mario; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; Sebastiani, Alessandro; Ghidini, Gabriella; Vedda, Anna; Fasoli, M.

被引用次数: 12

2009

Channel Hot-Carrier Degradation in Short-Channel Transistors With High-k/Metal Gate Stacks

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Amat, Esteve; Kauerauf, Thomas; Degraeve, Robin; De Keersgieter, An; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Groeseneken, Guido

被引用次数: 44

2009

Competing Degradation Mechanisms in Short-Channel Transistors Under Channel Hot-Carrier Stress at Elevated Temperatures

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Amat, Esteve; Kauerauf, Thomas; Degraeve, Robin; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Groeseneken, Guido

被引用次数: 40

2009

Time-Dependent Variability Related to BTI Effects in MOSFETs: Impact on CMOS Differential Amplifiers

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Martin-Martinez, Javier; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 17

2009

Understanding and Optimization of Hot-Carrier Reliability in Germanium-on-Silicon pMOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Maji, Debabrata; Crupi, Felice; Amat, Esteve; Simoen, Eddy; De Jaeger, Brice; Brunco, David P.; Manoj, C. R.; Rao, V. Ramgopal; Magnone, Paolo; Giusi, Gino; Pace, Calogero; Pantisano, Luigi; Mitard, Jerome; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 16

2009

Channel-Hot-Carrier Degradation and Bias Temperature Instabilities in CMOS Inverters

IEEE TRANSACTIONS ON ELECTRON DEVICES

Martin-Martinez, Javier; Gerardin, Simone; Amat, Esteve; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Paccagnella, Alessandro; Ghidini, Gabriella

被引用次数: 21

2009

Effects of the Localization of the Charge in Nanocrystal Memory Cells

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gasperin, Alberto; Amat, Esteve; Porti, Marc; Martin-Martinez, Javier; Nafria, Montserrat; Aymerich, Xavier; Paccagnella, Alessandro

被引用次数: 6

2009

Peculiar characteristics of nanocrystal memory cells programming window

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Gasperin, Alberto; Amat, Esteve; Martin, Javier; Porti, Marc; Nafria, Montserrat; Paccagnella, Alessandro

被引用次数: 0

2009

Gate oxide, wear-out and breakdown effects on the performance of analog and digital circuits

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fernandez, Raul; Martin-Martinez, J.; Rodriguez, R.; Nafria, Montserrat; Aymerich, Xavier H.

被引用次数: 38

2008

学术合作



Montserrat Nafria

巴塞罗那自治大学

合作次数: 49

Marc Porti

巴塞罗那自治大学

合作次数: 21

Javier Martin Martinez

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合作次数: 12

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北京大学

合作次数: 2

还有 10 个合作者